

WORKSHOP 4

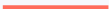
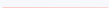
AI Course Assistant

Multimodal Two-Stage RAG & ReAct Agent System

LangGraph ReAct Agent • Pinecone Two-Stage Retrieval • Multimodal Vision • Tavily Web Search • Voice Output

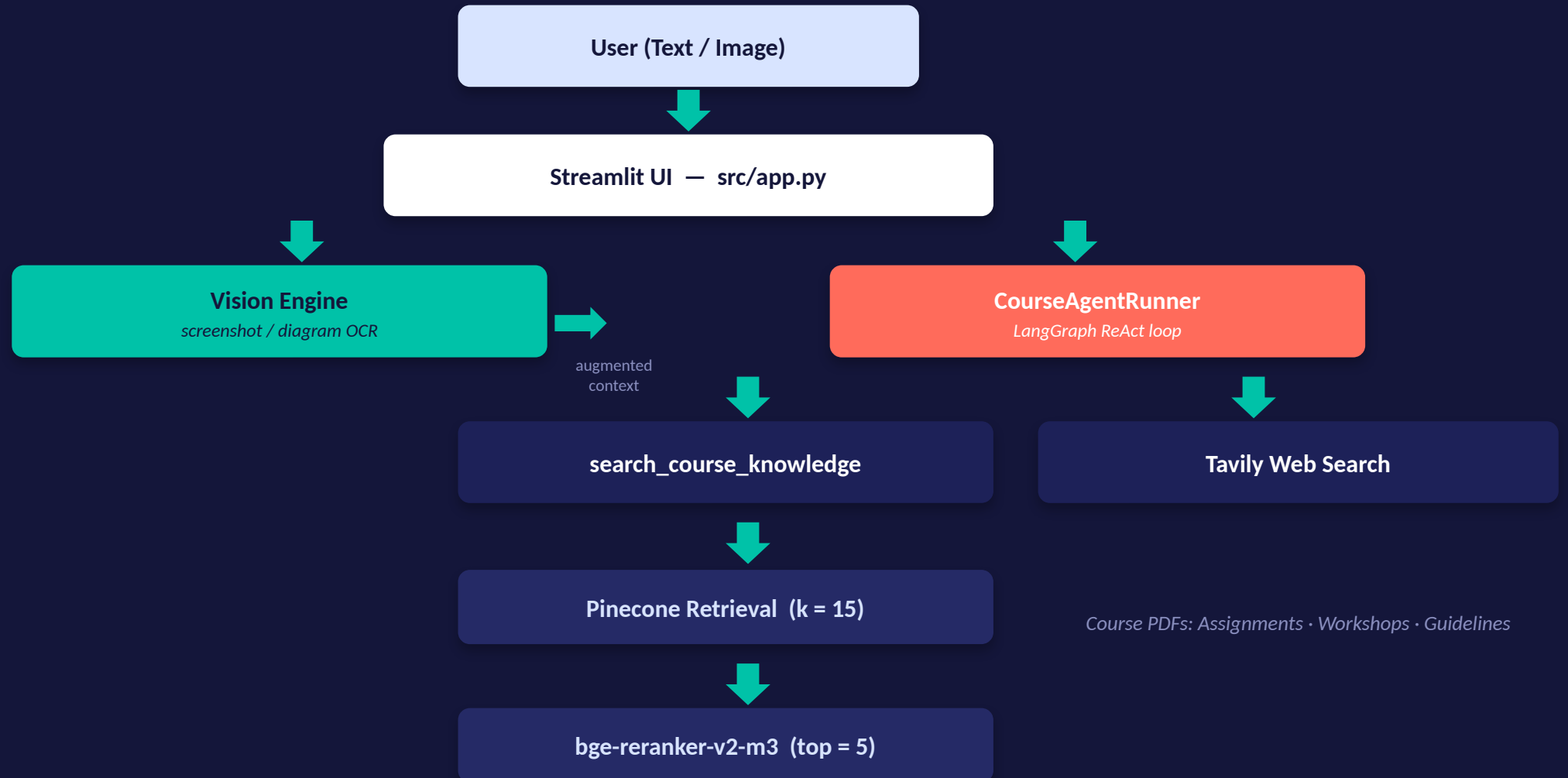
From Chatbot to Autonomous Agent

Workshop 3 answered questions. Workshop 4 reasons, retrieves, sees, and speaks.

	Vector Store	Local ChromaDB, basic ingestion		Pinecone Serverless + metadata filtering
	Retrieval	Single-stage dense search (top-5)		Two-stage: retrieve (k=15) → re-rank (top-5)
	Orchestration	Manual tool-calling loop		LangGraph ReAct Agent (autonomous)
	External Data	None — internal docs only		Live Tavily web search
	Input Modes	Text-only		Text + screenshot / diagram (Vision)

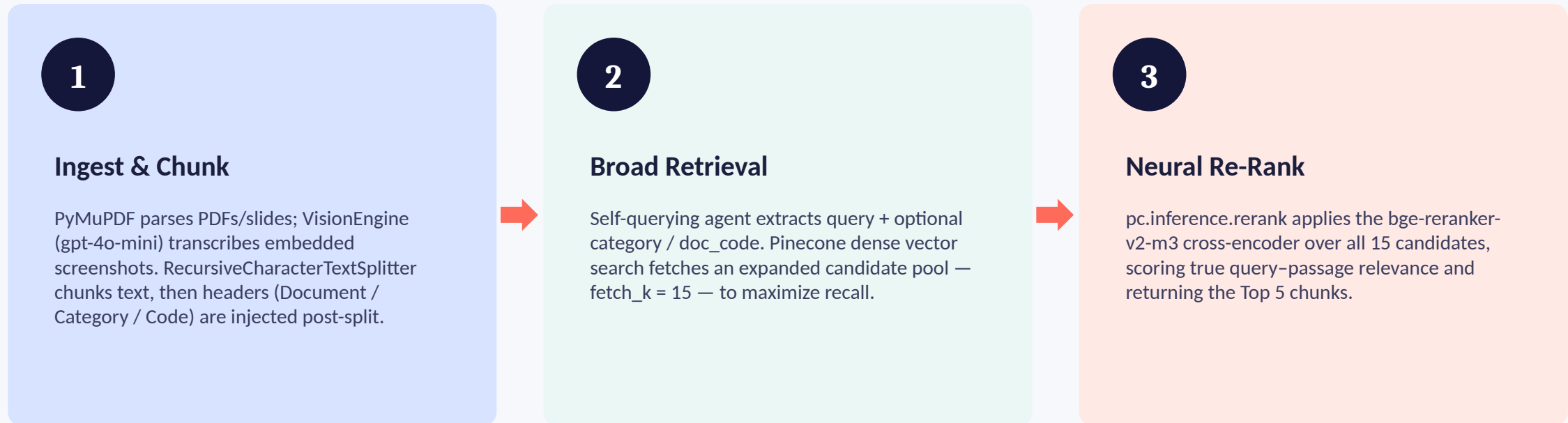
System Architecture

Streamlit UI → Vision / Agent → Tools → Knowledge Sources



Two-Stage Retrieval: Retrieve & Re-Rank

src/rag/vector_store.py — Pinecone Serverless + native inference re-ranking



Result: precise, cross-document-safe context — grounded in the exact chunks the LLM cross-attended to.

The ReAct Agent — Reasoning + Tools

src/agent/agent_runner.py + tools.py — LangGraph create_react_agent

1

Thought

Agent decides what info is missing and which tool can supply it.

2

Action

Calls search_course_knowledge (internal) or TavilySearch (external / live).

3

Observation

Reads tool output — retrieved chunks or search snippets.

4

Answer

Synthesizes a grounded reply and attaches source citations.

Available Tools

? search_course_knowledge

Self-querying: query (English), category (all / assignments / workshops / guidelines), doc_code (e.g. assignment_10). Returns Top-5 re-ranked chunks.

? TavilySearch

Real-time web search for library changes, external docs, current events not covered internally. Optional — enabled only if TAVILY_API_KEY is set.

? Bounded Sliding-Window Memory

Keeps the most recent 10 conversation turns — enough context for multi-turn follow-ups without runaway token cost.

Beyond Text: Vision & Voice

`src/engines/vision_engine.py` & `src/engines/tts_engine.py`



Multimodal Vision Engine

- Users upload a screenshot (error traceback, code, architecture diagram).
- gpt-4o-mini inspects the image with a Pydantic structured-output schema.
- Produces a clean text description — debugging notes or diagram summary.
- Description is appended to the user's prompt as augmented context before the agent runs.

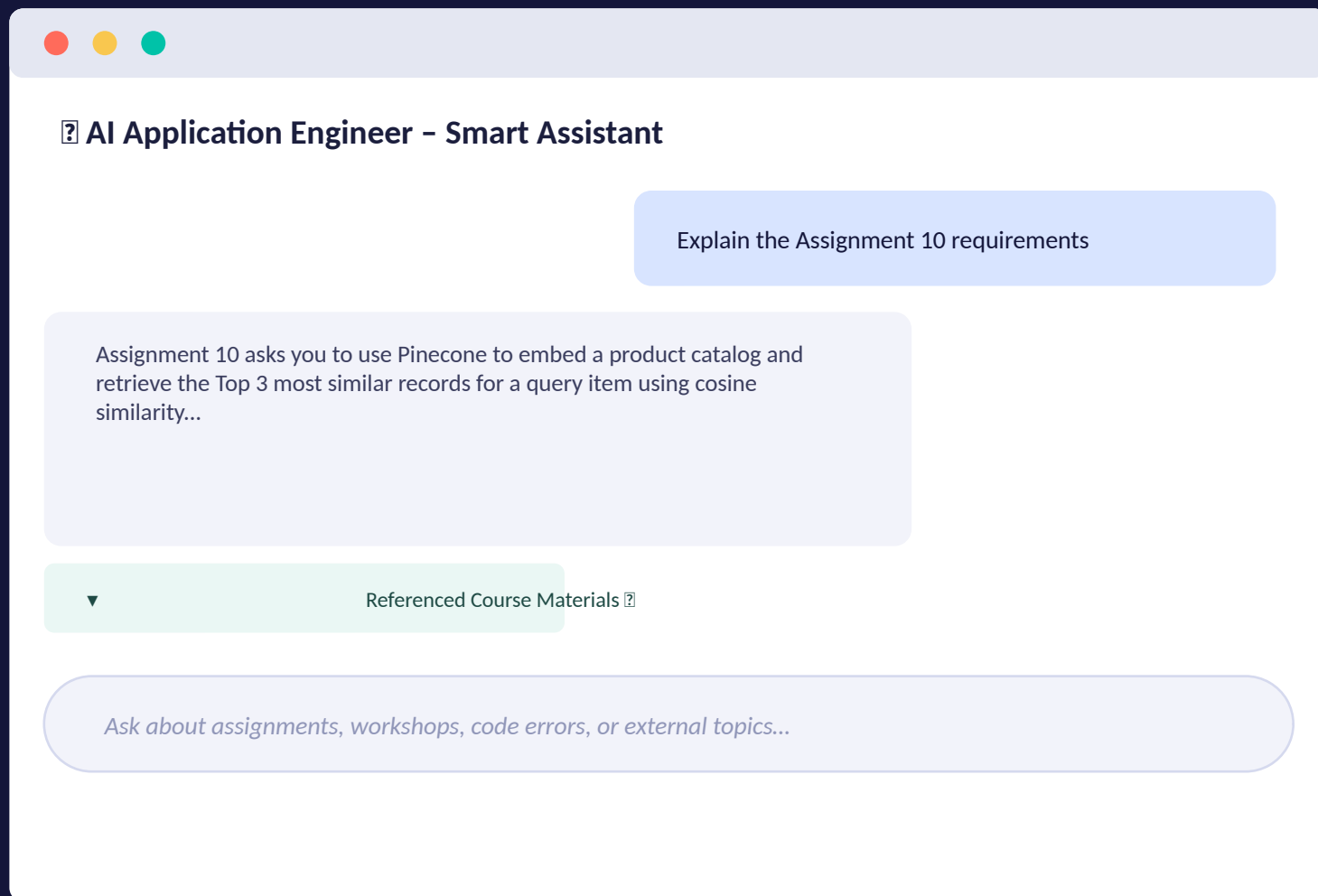


Text-to-Speech Output

- gTTS synthesizes the agent's final answer into spoken audio.
- langdetect auto-detects the response language (EN / VI) for natural pronunciation.
- Source filenames stay in the citation panel — never read aloud — for clean audio.
- Auto-play toggle in the sidebar lets presenters hear responses hands-free.

The Streamlit Experience

src/app.py — what the user actually sees



Multi-turn chat

Full session state with sliding-window memory



Image upload

Sidebar uploader for screenshots & diagrams



Source citations

Expandable panel with file + page numbers



Voice controls

Toggle TTS + auto-play from the sidebar



Live status

Shows Pinecone index & retrieval pipeline state

Technology Stack

requirements.txt — core dependencies powering the system

Orchestration

LangChain

core framework & chains

LangGraph

create_react_agent loop

langchain-tavily

web search tool

AI Models

gpt-4o-mini

chat, vision & embeddings

text-embedding-3-small

vector embeddings

bge-reranker-v2-m3

cross-encoder re-rank

Data & Retrieval

Pinecone Serverless

vector store + inference

PyMuPDF (fitz)

PDF / slide parsing

Pydantic

structured outputs

Experience

Streamlit

web UI

gTTS + langdetect

voice output

pytest + pytest-mock

unit & e2e tests

Live Demo Script

Four scenarios showing every capability end to end

1

Internal Knowledge

"What are the deliverables for Assignment 10?"

Agent calls `search_course_knowledge` → Pinecone retrieve (k=15) → re-rank (top=5)
→ cites source PDF + page.

2

Live Web Search

"What changed in the latest LangGraph release?"

Agent recognizes internal docs won't help → falls back to TavilySearch for current info.

3

Multimodal Vision

Upload a Python traceback screenshot

VisionEngine (gpt-4o-mini) reads the error → augments the prompt → agent explains the fix.

4

Multi-turn + Voice

Follow-up question referencing the previous answer

Sliding-window memory keeps last 10 turns; final answer is read aloud via gTTS.

Built by the Workshop 4 Team

Nine contributors, one autonomous course assistant

GiapHN

Vector store & document/OCR pipeline

CuongTQ7

LangGraph ReAct agent runner

DungPQ6

TTS + Vision engine

AnNPH5

Streamlit UI

LocNTH

Agent tools & self-querying

ThangTP3

Prompt design & demo questions

TrucNLT

Unit tests, e2e QA & conversation logs

VietNQ32

Architecture slides & presentation

ThanhVD7

End-to-end pipeline integration

Thank you — Questions & Live Demo